21-242 Recitation 12

Matrices — Q1. Definition of [T] and the map T [T]

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Define $[T]_{\mathcal{CB}}$ and show that $T \mapsto [T]_{\mathcal{CB}}$ is an invertible linear transformation.

Course-level solution

Definition. The *matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$* is the $m \times n$ matrix $[T]_{\mathcal{CB}}$ whose jth column is $[T(\vec{b}_j)]_{\mathcal{C}}$.

Let $\Psi : \mathcal{L}(V, W) \to \mathrm{Mat}_{m\times n}(\mathbb{F})$ by $\Psi(T) = [T]_{\mathcal{CB}}$.
Linearity. The jth column of $[S+T]_{\mathcal{CB}}$ is $[(S+T)(\vec{b}_j)]_{\mathcal{C}} = [S(\vec{b}_j)]_{\mathcal{C}} + [T(\vec{b}_j)]_{\mathcal{C}}$, and likewise $[aT]_{\mathcal{CB}} = a[T]_{\mathcal{CB}}$. So Ψ is linear.
Inverse. Given $A \in \mathrm{Mat}_{m\times n}(\mathbb{F})$ with columns $\vec{a}_1, \ldots, \vec{a}_n \in \mathbb{F}^m$, let $\vec{w}_j$ be the unique vector in W with $[\vec{w}_j]_{\mathcal{C}} = \vec{a}_j$, and let T be the unique linear map with $T(\vec{b}_j) = \vec{w}_j$ (Linear Transformations Q3). Then $[T]_{\mathcal{CB}} = A$. Uniqueness of T for a given matrix is the same construction in reverse. Thus Ψ is bijective, hence an invertible linear map: $\mathcal{L}(V, W) \cong \mathrm{Mat}_{m\times n}(\mathbb{F})$.

Later-course solution (using ideas from the rest of the handout)

Once dimensions are in hand, $\dim \mathcal{L}(V, W) = (\dim V)(\dim W) = nm = \dim \mathrm{Mat}_{m\times n}(\mathbb{F})$, so it is enough to check that Ψ is injective: $[T]_{\mathcal{CB}} = 0$ means $T(\vec{b}_j) = \vec{0}$ for all j, so $T = 0$.